6 Operational Semantics and Soundness

We formulate exactness for the *uniform fragment*: the syntax in Figure 5 without sequencing, non-strict comparisons, finite comparisons in source programs, or continuous samplers other than **`uniform`**. Write $Expr_\tau^{\mathsf{U}}$ for its well-typed expressions and $Expr_\tau^{\mathsf{U}+}$ for its operational closure under the finite comparisons and dispatch syntax emitted by Figure 13. The conditional claim is restricted to closed $e : \mathbf{bool}$ in $Expr_{\mathbf{bool}}^{\mathsf{U}}$:

If the emitted core term satisfies the measurable weak compiler-simulation, administrative-normalization, and terminal-limit compatibility obligations stated in the appendix, then $e : \mathbf{bool}$ and $\mathcal{D}[\![e : \mathbf{bool}]\!] : \mathbf{bool}$ are semantically equivalent.

The first step is to make formal sense of semantic equivalence through a monadic operational semantics . The remaining proof obligation is nontrivial: the emitted target takes deterministic administrative steps that have no matching source step, and substitution can change distribution dispatch. A proof therefore needs the weak compiler simulation described in §F; the earlier synchronous graph is insufficient.

6.1 Operational Semantics

The type-directedness of our discretization function $\mathcal{D}[\![\cdot]\!]$ requires our soundness argument to depend on the types of the given (sub-)expressions as well. We thus start by defining a *type-dependent semantic functional*, i.e., we define for every type τ a function $\langle\cdot\rangle : Expr_\tau \to \mathcal{D}(Vals_\tau + \bot)$, mapping each *fully type-annotated* expression $e : \tau$ to a *sub*-probability measure $\langle e : \tau \rangle$ over values of type τ or $\bot$. In the presence of possibly unbounded recursion *and* conditioning, our semantics has to distinguish between non-termination and observation failure, which we realize as follows: The "missing" probability mass in $\langle e : \tau \rangle$ is the probability that e diverges. The probability of $\bot$ is the probability of encountering an observation failure. With this model, semantic equivalence indeed captures our sought-after correctness claim: If $\langle e : \mathbf{bool} \rangle = \langle \mathcal{D}[\![e : \mathbf{bool}]\!] : \mathbf{bool} \rangle$, then both expressions have the same probability of producing true, false, or an observation failure.

6.1.1 Small Step Semantics. The type-dependent semantic functional $\langle\cdot\rangle$ is defined via a *small-step* semantic functional $[\![\cdot]\!] : Expr_\tau \to \mathcal{D}(Expr_\tau + \bot)$. Defining the latter requires us to carefully design a suitable measurable space over the set of type-annotated expressions of type τ and $\bot$. All details on these measure-theoretic constructions are presented in §F.2. Let us consider the key idea here.

Let $\tau = \mathbf{bool}$ for the sake of concreteness. Care must be taken since, e.g., sub-expressions of type $\mathbf{float}[B; \{3.5, 4.5\}]$ cannot evaluate to *arbitrary* reals but are restricted to the values $\{3.5, 4.5\}$. We tackle this by extracting from each type-annotated expression a suitable *skeleton*, where float constants are replaced by "holes" (hence the term "skeleton"). For instance, the expression

$$e \quad = \quad (x : \mathbf{float}[\{<3.5, <4.5\}; \top] \;<\; y : \mathbf{float}[\{<3.5, <4.5\}; \{3.5, 4.5\}]) : \mathbf{bool}$$

gives rise to the skeleton

$$s \quad = \quad (\square : \mathbf{float}[\{<3.5, <4.5\}; \top] \;<\; \square : \mathbf{float}[\{<3.5, <4.5\}; \{3.5, 4.5\}]) : \mathbf{bool}$$

and the type annotations tell us that the possible values these holes can be filled with are in $\mathbb{R} \times \{3.5, 4.5\}$. We can thus inherit a measurable space over all well-typed expressions arising from this particular skeleton s from the standard Borel measurable space over $\mathbb{R} \times \{3.5, 4.5\}$. The measurable space over *all* well-typed expressions of type $\mathbf{bool}$ (i.e., over $Expr_{\mathbf{bool}}$) is then obtained from the measurable spaces of *all* skeletons that give rise to expressions of type $\mathbf{bool}$.

Equipped with this family of measurable spaces, we define the small-step functional $[\![\cdot]\!]$ in the expected, type-preserving manner (cf. Tables 1 and 2). For instance, if e_1 is not a value, then[1]

$$[\![(\textbf{let}\ x = e_1 : \tau_1\ \textbf{in}\ e_2 : \tau_2) : \tau_2]\!] = \underbrace{[\![e_1 : \tau_1]\!] \gg\!= \lambda g \in Expr_{\tau_1}.\delta_{(\textbf{let}\ x=g\ \textbf{in}\ e_2:\tau_2):\tau_2}}$$ standard probabilistic monadic bind ove

i.e., $[\![e_1 : \tau_1]\!]$ yields the sub-distribution over expressions obtained from executing e_1 for *one step*, and the monadic bind $\gg\!=$ "folds" this distribution into the surrounding **let**-statement. The intuition for the remaining statements is analogous. Crucially, $\gg\!=$ propagates the probability of encountering $\bot$ — an observation failure — through executions. For instance, assume that e_1 from above is given by **observe** (false : **bool**) : **unit**. Since e_1 encounters an observation failure with probability 1, i.e., $[\![\textbf{observe}\ (\text{false} : \textbf{bool}) : \textbf{unit}]\!](\{\bot\}) = 1$, the same holds for the surrounding **let**-statement, i.e., also $[\![(\textbf{let}\ x = e_1 : \tau_1\ \textbf{in}\ e_2 : \tau_2) : \tau_2]\!](\{\bot\}) = 1$. The appendix proves the generic strict-bind results and records the remaining operational measurability obligations in §F.4.

6.1.2 From Small-Step to Big-Step Semantics. We now define $\langle\cdot\rangle$ via $[\![\cdot]\!]$. Put $S_\tau(e') = [\![e' : \tau]\!]$ for ordinary expressions $e' \in Expr_\tau$. We first lift the one-step semantics $[\![\cdot]\!]$ to an n-step semantics $\langle\cdot\rangle_n$ using the canonical extension $(S_\tau)_\bot$:

$$\langle e : \tau\rangle_0 = \delta_{e:\tau} \qquad \text{and} \qquad \langle e : \tau\rangle_{n+1}(A) = \int_{Expr_\tau+\bot} (S_\tau)_\bot(e')(A)\, \langle e : \tau\rangle_n(de').$$

Conditional on the operational measurability obligations stated in the appendix, this construction is well defined because $[\![\cdot]\!]$ is a sub-Markov kernel (Lemma F.40). The full finite-step measures on expressions are not increasing: a step moves mass away from its source expression. Following the terminal-limit construction in the appendix, let ν^e_n be the restriction of $\langle e : \tau\rangle_n$ to the measurable terminal subspace $Vals_\tau + \bot$. Terminal outcomes are absorbing, so $\nu^e_n(A) \le \nu^e_{n+1}(A)$ for every measurable terminal event A . We define

$$\langle e : \tau\rangle(A) = \lim_{n\to\infty} \nu^e_n(A).$$

The appendix proves eventwise monotonicity and countable additivity of this limit.

6.2 Soundness via Probabilistic Couplings

We state the result supplied by a completed weak simulation. The appendix records the exact missing obligation.

Proposition 6.1 (Conditional exactness). *Let $e : \textbf{bool} \in Expr^{\cup}_{\textbf{bool}}$ be closed. Assume the operational measurability obligations, a measurable weak target kernel, and a measurable weak coupling whose marginals are the source step and target weak step. Assume generated administrative reduction is normalizing, the restrictions of the coupling iterates to terminal related outcomes have an increasing limit, and the target projection of that limit is the ordinary target terminal semantics. If related terminal Boolean values agree and the coupling preserves observation-failure mass, then $\langle e : \textbf{bool}\rangle = \langle \mathcal{D}[\![e : \textbf{bool}$*

Under these terminal-limit compatibility premises, the limiting coupling has the two ordinary terminal semantics as its component pushforwards. Agreement of related terminal Booleans and preservation of observation-failure mass make the pushforwards agree on true, false, and $\bot$. The required weak simulation, administrative-normalization proof, and terminal-limit compatibility theorem have not yet been established for the literal emitted core term.

[1]Put formally, for a sub-Markov kernel $f : X \to \mathcal{D}(Y + \bot)$, $(\mu \gg\!= f)(A) = \int_{X+\bot} f_\bot(x)(A)\, \mu(dx)$, where $f_\bot(\bot) = \delta_\bot$; see Definition F.29 . δ_e denotes the Dirac distribution of e.

- **(Contains X):** $X = f^{-1}(Y) \in \Sigma_X$ since $Y \in \Sigma_Y$.
- **(Complements):** If $f^{-1}(A) \in \Sigma_X$, then $X \setminus f^{-1}(A) = f^{-1}(Y \setminus A) \in \Sigma_X$ since $Y \setminus A \in \Sigma_Y$.
- **(Countable unions):** If $f^{-1}(A_i) \in \Sigma_X$ for $i \in \mathbb{N}$, then $\bigcup_i f^{-1}(A_i) = f^{-1}(\bigcup_i A_i) \in \Sigma_X$ since $\bigcup_i A_i \in \Sigma_Y$.

Hence (X, Σ_X) is a measurable space.

Proof of ii) Two measurable spaces (X, Σ_X) and (Y, Σ_Y) are isomorphic iff there exists a bijection $f : X \to Y$ such that f and f^{-1} are measurable.

We first show that f is measurable. Measurability of f means that $f^{-1}(A) \in \Sigma_X$ for every $A \in \Sigma_Y$. Let $A \in \Sigma_Y$ be arbitrary. By definition of Σ_X,

$$f^{-1}(A) \in f^{-1}(\Sigma_Y) = \Sigma_X.$$

It follows that f is measurable.

Next we show that f^{-1} is measurable. Measurability of f^{-1} means that $(f^{-1})^{-1}(B) \in \Sigma_Y$ for every $B \in \Sigma_X$. Let $B \in \Sigma_X$ be arbitrary. By definition of Σ_X, there exists some $A \in \Sigma_Y$ such that $B = f^{-1}(A)$.

Observe that:

$$\begin{aligned} (f^{-1})^{-1}(B) &= f(B) && \text{(preimage of } f^{-1} \text{ is } f) \\ &= f(f^{-1}(A)) && \text{(substituting } B = f^{-1}(A)) \\ &= A && (f \text{ is a bijection}) \end{aligned}$$

Since $A \in \Sigma_Y$, it follows that f^{-1} is measurable.

Therefore f is a measurable bijection with measurable inverse, so it is an isomorphism of measurable spaces. □

Lemma F.4 (Minimality of generated σ-algebras). *Let X be a set and let $\mathcal{G} \subseteq \mathcal{P}(X)$. Suppose C is a σ-algebra on X such that $\mathcal{G} \subseteq C$. Then*

$$\sigma(\mathcal{G}) \subseteq C.$$

Proof. Recall that $\sigma(\mathcal{G})$ is defined as the smallest σ-algebra on X containing $\mathcal{G}$, given concretely by the intersection of all σ-algebras on X containing $\mathcal{G}$:

$$\sigma(\mathcal{G}) = \bigcap_{\substack{\mathcal{F} \text{ is a } \sigma\text{-algebra on } X \\ \mathcal{G} \subseteq \mathcal{F}}} \mathcal{F}.$$

Since C is a σ-algebra on X containing $\mathcal{G}$, it appears in this intersection. Therefore,

$$\sigma(\mathcal{G}) \subseteq C.$$

□

Definition F.5 (Sub-Markov kernels). Given measurable spaces (X, Σ_X) and (Y, Σ_Y), a sub-Markov kernel with source X and target Y is a function $P : X \times \Sigma_Y \to [0, 1]$ (or equivalently, $X \to \Sigma_Y \to [0, 1]$) that maps each source state $x \in X$ to a subprobability distribution over target states $P(x, -) : \Sigma_Y \to [0, 1]$.

We call P a *sub-Markov kernel* if it satisfies two conditions:

(1) *Subprobability measure:* For each source state $x \in X$, the map $A \mapsto P(x, A)$ must be a subprobability measure; in particular, $P(x, Y) \leq 1$.
(2) *Measurable function:* For each event $A \in \Sigma_Y$ in the target space, the map $x \mapsto P(x, A)$ must be a measurable function.

If additionally $P(x, Y) = 1$ for every $x \in X$, then P is a *Markov kernel*. Thus every Markov kernel is a sub-Markov kernel, but the converse need not hold.

Definition F.6 (Projection with error). Let $(X + \bot, \Sigma_{X+\bot})$ and $(Y + \bot, \Sigma_{Y+\bot})$ be measurable spaces, with (X, Σ_X) and (Y, Σ_Y) measurable spaces as well. Let $\mu \in \mathcal{D}((X \times Y) + \bot)$. We define the first and second projection measures as:

(1) $\pi_1(\mu)(A) := \mu(A \times Y)$ for every measurable $A \in \Sigma_X$, and $\pi_1(\mu)(\bot) := \mu(\bot)$
(2) $\pi_2(\mu)(B) := \mu(X \times B)$ for every measurable $B \in \Sigma_Y$, and $\pi_2(\mu)(\bot) := \mu(\bot)$

Then, $\pi_1(\mu)$ is a measure on $(X + \bot, \Sigma_{X+\bot})$ and $\pi_2(\mu)$ is a measure on $(Y + \bot, \Sigma_{Y+\bot})$.

F.2 Measure Space on Expressions

To make the probabilistic semantics precise, we construct an appropriate measure space structure on the set of expressions by factoring out real numbers using the notion of what we call *expression skeletons*.

Definition F.7 (Expression Skeleton). An expression skeleton s is a syntactic expression obtained from a program by replacing each occurrence of a float constant with a hole of the form $\square$. The set of skeletons is generated by the grammar:

$$
\begin{aligned}
s ::= {} & \square \mid x \mid \text{true} \mid \text{false} \mid k_{\#n} \mid \textbf{diverge} \mid () \\
& \mid \textbf{let}\ x = s_1\ \textbf{in}\ s_2 \mid \textbf{if}\ s_1\ \textbf{then}\ s_2\ \textbf{else}\ s_3 \\
& \mid s_1 < s_2 \mid s_1 \leq s_2 \mid s_1 <_{\#n} s_2 \mid s_1 \leq_{\#n} s_2 \\
& \mid \textbf{uniform}(s_1, s_2) \mid \textbf{gaussian}(s_1, s_2) \mid \textbf{exponential}(s) \mid \textbf{beta}(s_1, s_2) \mid \textbf{discrete}(p_0, \ldots, p_{n-1}) \\
& \mid s_1\ \&\&\ s_2 \mid s_1\ ||\ s_2 \mid \text{not } s \mid \textbf{observe}(s) \mid s_1; s_2 \\
& \mid (s_1, s_2) \mid \textbf{fst}\ s \mid \textbf{snd}\ s \\
& \mid \textbf{fun}\ x \to s \mid s_1\ s_2 \mid \textbf{fix}\ f\ x := s \\
& \mid \texttt{nil} \mid s_1 :: s_2 \\
& \mid (\textbf{match}\ s\ \textbf{with}\ \mid \texttt{nil} \to s_{\text{nil}} \mid h :: t \to s_{\text{cons}}\ \textbf{end})
\end{aligned}
$$

We denote by *Skel* the set of all expression skeletons, and by $holes(s)$ the number of distinct holes in skeleton s.

Example F.8. The expression $\textbf{uniform}(0.0, 1.0) < 0.5$ has the skeleton $\textbf{uniform}(\square, \square) < \square$ with three holes.

Definition F.9 (Filling a skeleton). Let $s \in Skel$ and let $\vec{r} = (r_1, \ldots, r_{holes(s)}) \in \mathbb{R}^{holes(s)}$. Define the expression $s[\vec{r}] \in Expr$ to be the result of replacing each hole $\square_i$ in s with the corresponding real constant r_i, for $1 \leq i \leq holes(s)$.

To recover a *typed* notion of skeletons, we extend the typing judgment to skeletons by adding a distinguished typing rule for holes.

Definition F.10 (Well-typed expression skeleton). Define a typing judgment for skeletons, written $\Gamma \vdash s : \tau$, with the understanding that all typing rules from expressions are preserved for skeletons, and that holes are typed as $\textbf{float}[B; V]$ according to the rule below:

$$\frac{}{\square : \textbf{float}[B; V]}$$

We denote by $Skel_\tau$ the set of all well-typed expression skeletons, where:

$$Skel_\tau := \{s \in Skel \mid s : \tau\}.$$

- If s is a value skeleton, then every expression in $Expr_{s;\tau}$ is a value, so $Vals_\tau \cap Expr_{s;\tau} = Expr_{s;\tau} \in \mathcal{F}_{Expr_{s;\tau}}$.
- If s is a non-value skeleton, then no expression in $Expr_{s;\tau}$ is a value, so $Vals_\tau \cap Expr_{s;\tau} = \emptyset \in \mathcal{F}_{Expr_{s;\tau}}$.

Thus, $Vals_\tau \cap Expr_{s;\tau} \in \mathcal{F}_{Expr_{s;\tau}}$ for any $s \in Skel_\tau$, so $Vals_\tau \in \mathcal{F}_{Expr_\tau} \subseteq \mathcal{F}_{Expr_\tau + \bot}$.

(2) **(Measurability of $\{\bot\}$):** Taking $A = \emptyset$ in Definition F.25 gives $\{\bot\} \in \mathcal{F}_{Expr_\tau + \bot}$.

(3) **(Measurability of N_τ):** $N_\tau = Expr_\tau \setminus Vals_\tau$ is the complement of $Vals_\tau$ in $Expr_\tau$. Since $Vals_\tau \in \mathcal{F}_{Expr_\tau}$ and $\mathcal{F}_{Expr_\tau}$ is a σ-algebra, $N_\tau \in \mathcal{F}_{Expr_\tau} \subseteq \mathcal{F}_{Expr_\tau + \bot}$.

□

Note that in general, given (X, Σ_X) and (Y, Σ_Y) to be measurable spaces, $((X \times Y) + \bot)$ forms a measurable space, since the product $(X \times Y)$ is measurable and for any measurable space (Z, Σ_Z), the space $(Z + \bot, \Sigma_{Z+\bot})$ with $\Sigma_{Z+\bot}$ generated by $\Sigma_Z \cup \bot$ is measurable.

F.3 Strict Sub-Markov Kleisli Composition with Error

Let X be a measurable space and equip $X + \bot$ with the sigma algebra generated by the measurable subsets of X and the singleton $\{\bot\}$. Write $\mathcal{D}(X + \bot)$ for the subprobability measures on this measurable space.

Definition F.28 ($\bot$-extension). Define the $\bot$-extension of $g : X \to \mathcal{D}(Y + \bot)$ to be $g_\bot : (X + \bot) \to \mathcal{D}(Y + \bot)$, obtained from g by:

$$g_\bot(e) = \begin{cases} \delta_\bot & \text{if } e = \bot \\ g(e) & \text{otherwise .} \end{cases}$$

Thus the added error point is absorbing. A continuation is specified only on ordinary inputs; its value at $\bot$ is not a free choice.

Definition F.29 (Strict sub-Markov Kleisli bind). For measurable spaces X, Y, define

$$\mathsf{return}_X : X \to \mathcal{D}(X + \bot), \qquad \mathsf{return}_X(x) = \delta_x,$$

$$\mathsf{bind}_{X,Y} : \mathcal{D}(X + \bot) \to \big(X \to \mathcal{D}(Y + \bot)\big) \to \mathcal{D}(Y + \bot),$$

$$\mathsf{bind}_{X,Y}(\mu)(g)(A) = \int_{X+\bot} g_\bot(x)(A)\, \mu(dx),$$

where g is required to be a sub-Markov kernel and A is measurable in $Y + \bot$. We write $\mu \gg\!= g$ only for this typed operation. By definition, it always integrates the canonical $\bot$-extension $g_\bot$, so

$$\delta_\bot \gg\!= g = \delta_\bot.$$

In particular, bind cannot recover from an error. By Lemma F.30, the integrand is a sub-Markov kernel, and the result is a subprobability measure because

$$(\mu \gg\!= g)(Y + \bot) = \int_{X+\bot} g_\bot(x)(Y + \bot)\, \mu(dx) \le \mu(X + \bot) \le 1.$$

Lemma F.30 ($\bot$-extension is a sub-Markov kernel). *Given a sub-Markov kernel $f : X \to \mathcal{D}(Y + \bot)$, its $\bot$ extension $f_\bot : (X + \bot) \to \mathcal{D}(Y + \bot)$ is also a sub-Markov kernel. If f is a Markov kernel, then $f_\bot$ is a Markov kernel.*

Proof. We verify the two conditions for $f_\bot$ to be a sub-Markov kernel.

(1) Subprobability measure: We show for each $e \in X + \bot$, the map

$$A \mapsto f_\bot(e)(A)$$

is a subprobability measure.

Fix $e \in X + \bot$.

- If $e = \bot$, then $f_\bot(\bot) = \delta_\bot$, the Dirac measure at $\bot$, which is a probability measure on $Y + \bot$.
- If $e \in X$, then $f_\bot(e) = f(e)$, and since f is a sub-Markov kernel by assumption, $f(e)$ is a subprobability measure on $Y + \bot$.

Thus, for all $e \in X + \bot$, the map $A \mapsto f_\bot(e)(A)$ is a subprobability measure.

(2) Measurable function: We show for each $A \in \mathcal{F}_{Y+\bot}$, the map $e \mapsto f_\bot(e)(A)$ is a measurable function.

Fix $A \in \mathcal{F}_{Y+\bot}$. Define

$$g\colon (X + \bot) \to [0,1], \qquad g(e) := f_\bot(e)(A).$$

We prove g is measurable by showing $g^{-1}(B) \in \mathcal{F}_{X+\bot}$ for every $B \in \mathcal{B}([0,1])$. It suffices to prove that for every $a \in \mathbb{R}$,

$$g^{-1}((a,\infty) \cap [0,1]) \in \mathcal{F}_{X+\bot}$$

because the family of sets $\{(a,\infty) \cap [0,1] : a \in \mathbb{R}\}$ generates $\mathcal{B}([0,1])$.

Because $g(e) \in [0,1]$ for all $e \in X + \bot$ by definition, we have for every $a \in \mathbb{R}$,

$$\begin{aligned} g^{-1}\big((a,\infty) \cap [0,1]\big) &= \{e \in X + \bot \mid g(e) \in (a,\infty) \cap [0,1]\} \\ &= \{e \in X + \bot \mid (g(e) \in (a,\infty)) \wedge (g(e) \in [0,1])\} \\ &= \{e \in X + \bot \mid (g(e) > a) \wedge (g(e) \in [0,1])\} \\ &= \{e \in X + \bot \mid g(e) > a\}, \end{aligned}$$

where the last step uses that $g(e) \in [0,1]$ always holds. Hence it suffices to show that $\{e \in X + \bot \mid g(e) > a\} \in \mathcal{F}_{X+\bot}$ for every $a \in \mathbb{R}$.

- If $a \geq 1$: $\{e \in X + \bot \mid g(e) > a\} = \emptyset \in \mathcal{F}_{X+\bot}$.
- If $a < 0$: $\{e \in X + \bot \mid g(e) > a\} = X + \bot \in \mathcal{F}_{X+\bot}$.
- If $0 \leq a < 1$: $\{e \in X + \bot \mid g(e) > a\}$.

The only nontrivial case is the last. It splits into two sets:

$$\begin{aligned} \{e \in X + \bot \mid g(e) > a\} &= \{e \in X + \bot \mid f_\bot(e)(A) > a\} \\ &= \{e \in X \mid f_\bot(e)(A) > a\} \cup \{\bot \mid f_\bot(e)(A) > a\} \\ &= \{e \in X \mid f(e)(A) > a\} \cup \{\bot \mid \delta_\bot(A) > a\} \end{aligned}$$

The first set is measurable in X (and hence $X + \bot$) since f is a sub-Markov kernel by assumption. The second set is either $\emptyset \in \mathcal{F}_{X+\bot}$ or $\{\bot\} \in \mathcal{F}_{X+\bot}$. Since $\mathcal{F}_{X+\bot}$ is closed under countable unions, then $g^{-1}(B) \in \mathcal{F}_{X+\bot}$ for every $B \in \mathcal{B}([0,1])$. Thus, for all $A \in \mathcal{F}_{Y+\bot}$, the map $e \mapsto f_\bot(e)(A)$ is a measurable function.

Since both conditions hold, $f_\bot$ is a sub-Markov kernel. If f is Markov, both cases above have total mass one, so $f_\bot$ is Markov. □

Lemma F.31 (Dirac return is a Markov kernel). *For any measurable space $(X, \mathcal{F}_X)$, the map*

$$\mathsf{return} : X \to \mathcal{D}(X + \bot), \qquad x \mapsto \delta_x.$$

is a Markov kernel (and hence a sub-Markov kernel).

Proof. We verify the two kernel conditions and unit total mass.

(1) Probability measure: We show for each $x \in X$, the map

$$A \mapsto \mathsf{return}(x)(A) = \delta_x(A)$$

is a probability measure.

Fix $x \in X$. δ_x is the Dirac measure at x, hence a probability measure on $X+\bot$, with $\delta_x(X + \bot) = 1$.

(2) Measurable function: We show that for each $A \in \mathcal{F}_{X+\bot}$, the map

$$x \mapsto \mathsf{return}(x)(A) = \delta_x(A)$$

is a measurable function.

Fix $A \in \mathcal{F}_{X+\bot}$. Define

$$g : X \to [0,1], \qquad g(x) := \delta_x(A).$$

We prove g is measurable by showing $g^{-1}(B) \in \mathcal{F}_X$ for every $B \in \mathcal{B}([0,1])$. It suffices to prove that for every $a \in \mathbb{R}$,

$$g^{-1}\big((a,\infty) \cap [0,1]\big) \in \mathcal{F}_X,$$

because the family $\{(a,\infty) \cap [0,1] : a \in \mathbb{R}\}$ generates $\mathcal{B}([0,1])$.

Because $g(e) \in [0,1]$ for all $e \in X + \bot$, we have for every $a \in \mathbb{R}$,

$$\begin{aligned} g^{-1}\big((a,\infty) \cap [0,1]\big) &= \{e \in X \mid g(e) \in (a,\infty) \cap [0,1]\} \\ &= \{e \in X \mid (g(e) \in (a,\infty)) \wedge (g(e) \in [0,1])\} \\ &= \{e \in X \mid (g(e) > a) \wedge (g(e) \in [0,1])\} \\ &= \{e \in X \mid g(e) > a\}, \end{aligned}$$

where the last step uses that $g(e) \in [0,1]$ always holds. Hence it suffices to show that $\{e \in X \mid g(e) > a\} \in \mathcal{F}_X$ for every $a \in \mathbb{R}$.

- If $a \geq 1$: $\{x \in X \mid g(x) > a\} = \emptyset \in \mathcal{F}_X$.
- If $a < 0$: $\{x \in X \mid g(x) > a\} = X \in \mathcal{F}_X$.
- If $0 \leq a < 1$: $\{x \in X \mid g(x) > a\}$.

The only nontrivial case is the last. Using $g(x) = \delta_x(A)$ and the fact that $\delta_x(A) \in \{0,1\}$ for all $x \in X$, we have for $0 \leq a < 1$:

$$\begin{aligned} \{x \in X \mid g(x) > a\} &= \{x \in X \mid \delta_x(A) > a\} \\ &= \{x \in X \mid \delta_x(A) = 1\} \\ &= \{x \in X \mid x \in A\} \\ &= A \cap X. \end{aligned}$$

Since $A \in \mathcal{F}_{X+\bot}$ and $\mathcal{F}_X = \{X \cap A \mid A \in \mathcal{F}_{X+\bot}\}$ is the subspace σ-algebra on $X \subseteq X + \bot$, it follows that $A \cap X \in \mathcal{F}_X$.

Therefore $\{x \in X \mid g(x) > a\} \in \mathcal{F}_X$ for all $a \in \mathbb{R}$, so $g^{-1}\big((a,\infty) \cap [0,1]\big) \in \mathcal{F}_X$ for all a. Hence $g^{-1}(B) \in \mathcal{F}_X$ for every $B \in \mathcal{B}([0,1])$, and g is measurable. □

Lemma F.32 (Zero kernel is a sub-Markov kernel). *Let $(X, \mathcal{F}_X)$ and $(Y, \mathcal{F}_Y)$ be measurable spaces. The constant-zero map*

$$f : X \to \mathcal{D}(Y), \qquad f(x)(A) := 0$$

is a sub-Markov kernel.

Proof. We verify the two kernel conditions.

(1) Subprobability measure. We show for each $x \in X$, the map

$$A \mapsto f(x)(A)$$

is a subprobability measure.

Fix any $x \in X$. Then $f(x)$ is the zero measure: it satisfies σ-additivity since

$$f(x)\left(\bigcup_i A_i\right) = 0 = \sum_i 0 = \sum_i f(x)(A_i)$$

for every pairwise disjoint family $(A_i)_i$, and total mass $f(x)(Y) = 0 \leq 1$.

(2) Measurable function. For any $A \in \mathcal{F}_Y$, the map $x \mapsto f(x)(A) = 0$ is the constant-zero function on X, hence measurable.

Thus both conditions hold, so f is a sub-Markov kernel. □

Lemma F.33 (Composition of sub-Markov kernels). *Let (X, Σ_X), (Y, Σ_Y), and (Z, Σ_Z) be measurable spaces. If*

$$P : X \to \mathcal{D}(Y) \quad \text{and} \quad Q : Y \to \mathcal{D}(Z)$$

are sub-Markov kernels, then their composition

$$(P \gg\!= Q)(x, A) \;:=\; \int_Y Q(y, A)\, P(x, dy)$$

defines a sub-Markov kernel

$$P \gg\!= Q : X \to \mathcal{D}(Z).$$

Proof. For each $x \in X$, the map $A \mapsto (P \gg\!= Q)(x, A)$ is a measure by the standard integral construction. Its total mass is bounded by

$$(P \gg\!= Q)(x, Z) = \int_Y Q(y, Z)\, P(x, dy) \leq \int_Y 1\, P(x, dy) = P(x, Y) \leq 1.$$

For each $A \in \Sigma_Z$, measurability of $y \mapsto Q(y, A)$ and the standard measurability theorem for integration against a kernel imply that $x \mapsto \int_Y Q(y, A)\, P(x, dy)$ is measurable; see [30, Lemma 1.38]. Hence the composition is a sub-Markov kernel. □

Lemma F.34 (Typed strict Kleisli laws). *Let*

$$g : X \to \mathcal{D}(Y + \bot) \quad \text{and} \quad h : Y \to \mathcal{D}(Z + \bot)$$

be sub-Markov kernels, and let $\mu \in \mathcal{D}(X + \bot)$. Then the following equalities of measures are well typed:

$$\delta_x \gg\!= g = g(x) \qquad (x \in X), \tag{1}$$

$$\mu \gg\!= \mathsf{return}_X = \mu, \tag{2}$$

$$(\mu \gg\!= g) \gg\!= h = \mu \gg\!= \big(\lambda x \in X.\, g(x) \gg\!= h\big). \tag{3}$$

Moreover, $\delta_\bot \gg\!= g = \delta_\bot$.

Proof. The left-unit and strictness equations follow by integration against a Dirac measure:

$$(\delta_x \gg\!= g)(A) = g_\perp(x)(A) = g(x)(A), \qquad (\delta_\perp \gg\!= g)(A) = g_\perp(\perp)(A) = \delta_\perp(A).$$

The $\perp$-extension of return_X is the identity Dirac kernel $u \mapsto \delta_u$ on $X + \perp$. Hence

$$(\mu \gg\!= \mathrm{return}_X)(A) = \int_{X+\perp} \delta_u(A)\, \mu(du) = \mu(A).$$

For associativity, define the ordinary-input kernel

$$k : X \to \mathcal{D}(Z + \perp), \qquad k(x) = g(x) \gg\!= h.$$

It is a sub-Markov kernel by Lemmas F.30 and F.33. For measurable $A \subseteq Z + \perp$, kernel integration gives

$$\begin{aligned}((\mu \gg\!= g) \gg\!= h)(A) &= \int_{X+\perp} \left(\int_{Y+\perp} h_\perp(y)(A)\, g_\perp(x, dy) \right) \mu(dx) \\ &= \int_{X+\perp} k_\perp(x)(A)\, \mu(dx) \\ &= (\mu \gg\!= k)(A).\end{aligned}$$

The middle equality holds for $x \in X$ by the definition of k. At $x = \perp$, its left side is $h_\perp(\perp)(A) = \delta_\perp(A)$, which is exactly $k_\perp(\perp)(A)$. □

F.4 Small-Step Semantics

Throughout this section, $e : \tau$ denotes a well-typed expression whose subexpressions are fully type annotated. The dynamic-parameter transformation in Figure 13 denotes the displayed nest of core-language let-bindings , conditionals, Boolean conjunctions, finite equalities, and a discrete choice. No auxiliary one-step dispatch constructor is added to the language. The historical synchronous coupling below is disabled because its proof-only $\mathtt{DC}_{B,B_1,B_2,V_1,V_2}$ constructor does not describe the literal emitted target.

Equipped with these notions, we define the small-step semantics of our language:

Definition F.35. The small-step semantics $[\![e : \tau]\!] \in \mathcal{D}(Expr_\tau + \perp)$ is defined recursively on the structure of $e : \tau$ by the rules in Table 1 and Table 2.

The missing probability mass in $[\![e : \tau]\!]$ is the probability of encountering an error due to ill-formed parameters fed into distributions (e.g., **uniform**$(0.8, 0.1)$). $[\![e : \tau]\!](\perp)$ is the probability of encountering an observation failure.

Lemma F.36 (Measurability of constant extension map). *Let $A \subseteq \mathbb{R}^n$ be equipped with the subspace σ-algebra*

$$\mathcal{F}_A := \{A \cap M \mid M \in \mathcal{B}(\mathbb{R}^n)\},$$

and let $B \subseteq \mathbb{R}^d$ be equipped with the subspace σ-algebra

$$\mathcal{F}_B := \{B \cap N \mid N \in \mathcal{B}(\mathbb{R}^d)\}.$$

Fix constant vectors $c_1 \in \mathbb{R}^{m_1}, \ldots, c_k \in \mathbb{R}^{m_k}$. Define

$$g : \mathbb{R}^n \to \mathbb{R}^d$$

to be any map obtained by concatenating blocks consisting of

- the holes coming from the float constants of e itself.

By Lemma F.36, ψ_s is measurable.
By construction, for all $\vec{r} \in A_{s;\tau_1}$:

$$e[s[\vec{r}]/x] = t[\psi_s(\vec{r})].$$

Define the measurable set of hole-assignments corresponding to A at skeleton t:

$$\begin{aligned} A_t :=& \{\vec{u} \in A_{t;\tau_2} \mid t[\vec{u}] \in A\} \\ :=& \{\vec{u} \in A_{t;\tau_2} \mid t[\vec{u}] \in A \cap Expr_{t;\tau_2}\} \qquad (t[\vec{u}] \text{ ranges over } Expr_{t;\tau_2}) \end{aligned}$$

By Definition F.23, $A \cap Expr_{t;\tau_2} \in \mathcal{F}_{Expr_{t;\tau_2}}$, and by Lemma F.3, $A_t \in \mathcal{F}_{t;\tau_2}$. We compute:

$$\begin{aligned} R_s &= \{\vec{r} \in A_{s;\tau_1} \mid e[s[\vec{r}]/x] \in A\} \\ &= \{\vec{r} \in A_{s;\tau_1} \mid t[\psi_s(\vec{r})] \in A\} \\ &= \{\vec{r} \in A_{s;\tau_1} \mid \psi_s(\vec{r}) \in A_t\} \\ &= \psi_s^{-1}(A_t). \end{aligned}$$

Since ψ_s is measurable and $A_t \in \mathcal{F}_{t;\tau_2}$, we have $R_s = \psi_s^{-1}(A_t) \in \mathcal{F}_{s;\tau_1}$.
Since $R_s \in \mathcal{F}_{s;\tau_1}$ for every $s \in Skel_{\tau_1}$, we get $C \in \mathcal{F}_{Expr_{\tau_1}}$, and therefore

$$\mathrm{Sub}_{e,x}^{-1}(A) = C \cap Val_{\tau_1} \in \mathcal{F}_{Val_{\tau_1}}.$$

Hence $\mathrm{Sub}_{e,x}$ is measurable. □

Operational measurability obligations (open). The following kernel results are conditional on three facts for every rule in Tables 1 and 2: the syntactic case domains and side-condition regions form measurable partitions; every deterministic reconstruction or substitution map used under a Dirac measure is measurable; and every primitive sampling clause is a sub-Markov kernel on its parameter-validity region. The preceding lemmas establish representative substitution and reconstruction cases, but they do not discharge these obligations for every rule.

Lemma F.39 (Strict bind continuations, conditional). *Assuming the operational measurability obligations above, every continuation in a bind $\mu \gg\!= f$ in the small-step rules is a sub-Markov kernel*

$$f : Expr_{\tau_1} \to \mathcal{D}(Expr_{\tau_2} + \bot).$$

The operational bind uses its canonical extension $f_\bot : (Expr_{\tau_1} + \bot) \to \mathcal{D}(Expr_{\tau_2} + \bot)$.

Proof. We first prove that $f : Expr_{\tau_1} \to \mathcal{D}(Expr_{\tau_2} + \bot)$ is a sub-Markov kernel. Let e be an expression whose evaluation rule contains a monadic bind $\mu \gg\!= f$. We proceed by case analysis on e:

- Consider $e = (\mathbf{let}\ y = x : \tau_1\ \mathbf{in}\ e_2 : \tau_2) : \tau_2$.
  This means proving that $\lambda x.\, \delta_{\mathbf{let}\ y=x\ \mathbf{in}\ e_2}$ is a Markov kernel (and hence a sub-Markov kernel).
  Formally, define

  $$f\colon Expr_{\tau_1} \to \mathcal{D}(Expr_{\tau_2} + \bot), \qquad f(x) = \delta_{\mathbf{let}\ y=x\ \mathbf{in}\ e_2}.$$

  We verify the two conditions for f to be a Markov kernel.

Define

$$R_s := \{\vec{r} \in A_{s;\tau_1} \mid \textbf{let } y = s[\vec{r}] \textbf{ in } e_2 \in A\}.$$

Let s_2 be the skeleton of e_2, with $m := holes(s_2)$, and let $\vec{c} \in A_{s_2;\tau_2} \subseteq \mathbb{R}^m$ be the unique tuple such that $e_2 = s_2[\vec{c}]$. Define the combined skeleton

$$s' := \textbf{let } y = s \textbf{ in } s_2 \in Skel_{\tau_2},$$

whose holes consist of those from s (filled by $\vec{r} \in \mathbb{R}^n$) followed by those from s_2 (filled by $\vec{c} \in \mathbb{R}^m$), so that $A_{s';\tau_2} \subseteq \mathbb{R}^{n+m}$. Thus, for every $\vec{r} \in A_{s;\tau_1}$:

$$\textbf{let } y = s[\vec{r}] \textbf{ in } e_2 = s'[(\vec{r}, \vec{c})].$$

Define a constant-extension map

$$\phi_{\vec{c}} \colon A_{s;\tau_1} \to A_{s';\tau_2}, \qquad \phi_{\vec{c}}(\vec{r}) := (\vec{r}, \vec{c}).$$

By Lemma F.36, $\phi_{\vec{c}}$ is measurable.
Define the measurable set of hole-assignments corresponding to A at skeleton s':

$$\begin{aligned} A_{s'} :=& \{(\vec{r}, \vec{r}\,') \in A_{s';\tau_2} \mid s'[\vec{r}, \vec{r}\,'] \in A\} \\ :=& \{(\vec{r}, \vec{r}\,') \in A_{s';\tau_2} \mid s'[\vec{r}, \vec{r}\,'] \in A \cap Expr_{\tau_2}\} && \text{(Definition F.1)} \\ :=& \{(\vec{r}, \vec{r}\,') \in A_{s';\tau_2} \mid s'[\vec{r}, \vec{r}\,'] \in A \cap Expr_{s';\tau_2}\} && (s'[\vec{r}, \vec{r}\,'] \text{ ranges over } Expr_{s';\tau_2}) \end{aligned}$$

Since $A \cap Expr_{s';\tau_2} \in \mathcal{F}_{Expr_{s';\tau_2}}$, by Lemma F.21, $A_{s'} \in \mathcal{F}_{s';\tau_2}$.

Then:

$$\begin{aligned} R_s &= \{\vec{r} \in A_{s;\tau_1} \mid \textbf{let } y = s[\vec{r}] \textbf{ in } e_2 \in A\} \\ &= \{\vec{r} \in A_{s;\tau_1} \mid s'[(\vec{r}, \vec{c})] \in A\} \\ &= \{\vec{r} \in A_{s;\tau_1} \mid s'[\phi_{\vec{c}}(\vec{r})] \in A\} \\ &= \{\vec{r} \in A_{s;\tau_1} \mid \phi_{\vec{c}}(\vec{r}) \in A_{s'}\} \\ &= \phi_{\vec{c}}^{-1}(A_{s'}). \end{aligned}$$

Since $A_{s'} \in \mathcal{F}_{s';\tau_2}$ and $\phi_{\vec{c}}$ is measurable, we conclude

$$\phi_{\vec{c}}^{-1}(A_{s'}) \in \mathcal{F}_{s;\tau_1},$$

and therefore $B \cap Expr_{s;\tau_1} \in \mathcal{F}_{Expr_{s;\tau_1}}$ for all $s \in Skel_{\tau_1}$, giving $B \in \mathcal{F}_{Expr_{\tau_1}}$ as required.

Hence, $\{x \in Expr_{\tau_1} \mid g(x) > a\} \in \mathcal{F}_{Expr_{\tau_1}}$ for all $a \in \mathbb{R}$, so $g^{-1}((a, \infty) \cap [0, 1]) \in \mathcal{F}_{Expr_{\tau_1}}$ for all $a \in \mathbb{R}$. Hence $g^{-1}(B) \in \mathcal{F}_{Expr_{\tau_1}}$ for every $B \in \mathcal{B}([0, 1])$, and g is measurable.

Thus, $f \colon Expr_{\tau_1} \to \mathcal{D}(Expr_{\tau_2} + \bot)$ is a Markov kernel, and hence a sub-Markov kernel.

- For every remaining rule, measurability of its case domain, reconstruction maps, and primitive kernels is one of the stated operational measurability obligations. Those obligations imply the required ordinary-input continuation is a sub-Markov kernel.

By Lemma F.30, the canonical $\bot$-extension $f_\bot$ is a sub-Markov kernel and is the kernel integrated by the strict bind. □

Lemma F.40 (Small-step kernel, conditional). *Assuming the operational measurability obligations above, for each type τ the small-step semantics*

$$[\![\cdot]\!] : Expr_\tau \to \mathcal{D}(Expr_\tau + \bot)$$

is a sub-Markov kernel.

Proof. Proceed by structural induction through the operational rules. On each measurable case domain, deterministic clauses are Dirac kernels induced by the assumed measurable reconstruction map, primitive clauses are the assumed sub-Markov kernels, and divergence is the zero kernel from Lemma F.32. A bind clause first uses the induction hypothesis and then composes it with the canonical extension $f_\perp$ of the ordinary-input continuation supplied by Lemma F.39. This composition is a sub-Markov kernel by Lemmas F.30 and F.33. Combining the branch kernels over the assumed measurable partition proves the claim. □

F.5 Small-Step Coupling Semantics

Weak compiler-simulation obligation (open). Let $\longrightarrow_a$ contain only deterministic administrative target reductions: let substitution, finite equality, Boolean conjunction, and conditional selection. Write $\Longrightarrow_a$ for its reflexive transitive closure. A sound proof for the literal generated term must construct a measurable relation R_τ containing $(e, \mathcal{D}[\![e]\!])$ and closed under a source probabilistic step matched by finitely many target administrative steps, one target probabilistic step, and finitely many further administrative steps. The symmetric clause must permit target administrative steps while the source stutters. Related terminal Boolean values must be equal, and both sides must assign the same mass to observation failure. Establishing this weak simulation remains open.

Even such a weak relation is not by itself a terminal theorem. The conditional result below separately assumes that generated administrative reduction is normalizing and that iteration of the measurable target weak-step kernel has the same terminal limit as iteration of the ordinary target small-step kernel.

F.6 Big-Step Semantics

Define the ordinary-input kernel

$$S_\tau(e') = [\![e' : \tau]\!].$$

We extend the ordinary small-step semantics to an n-step semantics by explicitly integrating the canonical bottom-preserving extension:

$$\langle e : \tau\rangle_n = \begin{cases} \delta_{e:\tau} & \text{if } n = 0 \\ \lambda A.\ \displaystyle\int_{Expr_\tau + \perp} (S_\tau)_\perp(e')(A)\ \langle e : \tau\rangle_{n-1}(de') & \text{if } n > 0\ . \end{cases}$$

As a sanity check, we show that the n-step semantics defines a sub-Markov kernel.

Lemma F.41 (n-step kernel, conditional). *Under the operational measurability obligations, for all $n \in \mathbb{N}$, $\langle\cdot\rangle_n : Expr_\tau \to \mathcal{D}(Expr_\tau + \perp)$ is a sub-Markov kernel.*

Proof. By induction on n.

Base case: $n = 0$. By definition,

$$\langle e : \tau\rangle_n = \delta_{e:\tau}$$

i.e. the map $e \mapsto \delta_e$. By Lemma F.31, this is a Markov kernel, and hence a sub-Markov kernel.

Inductive step: $n > 0$. By definition,

$$\langle e : \tau\rangle_n = \langle e : \tau\rangle_{n-1} \gg\!= \lambda e'.[\![e' : \tau]\!]$$

By the I.H., $\langle\cdot\rangle_{n-1} : Expr_\tau \to \mathcal{D}(Expr_\tau + \perp)$ is a sub-Markov kernel. By Lemma F.40, $\lambda e'.[\![e' : \tau]\!] : Expr_\tau \to \mathcal{D}(Expr_\tau + \perp)$ is a sub-Markov kernel. By Lemma F.30, its $\perp$-extension from $(Expr_\tau + \perp)$ to $\mathcal{D}(Expr_\tau + \perp)$ is also a sub-Markov kernel. By Lemma F.33, $\langle e : \tau\rangle_{n-1} \gg\!= \lambda e'.[\![e' : \tau]\!]$ is a sub-Markov kernel. □

For every type τ, let $T_\tau = Vals_\tau + \bot$ be the measurable subspace of terminal outcomes (cf. Lemma F.27.1). For $e : \tau$, define the restriction of its n-step semantics to terminal outcomes by

$$\nu_n^e(A) = \langle e : \tau \rangle_n(A) \qquad (A \in \mathcal{F}_{T_\tau}).$$

Each ν_n^e is a subprobability measure. The full n-step measures on expressions need not be increasing: a deterministic step removes mass from the singleton containing its source expression.

Lemma F.42 (Increasing terminal restrictions). *For every $e : \tau$, $n \in \mathbb{N}$, and measurable terminal event $A \in \mathcal{F}_{T_\tau}$,*

$$\nu_n^e(A) \le \nu_{n+1}^e(A).$$

Proof. Let $K_\tau = (S_\tau)_\bot$ be the bottom-extended one-step kernel. Terminal outcomes are absorbing: $K_\tau(x) = \delta_x$ for $x \in T_\tau$. Thus, for every $x \in Expr_\tau + \bot$ and measurable $A \subseteq T_\tau$, $K_\tau(x)(A) \ge \mathbf{1}_A(x)$. Integrating this inequality against $\langle e : \tau \rangle_n$ gives

$$\nu_{n+1}^e(A) = \int K_\tau(x)(A)\, \langle e : \tau \rangle_n(dx) \ge \nu_n^e(A).$$

□

Definition F.43 (Big-step semantics). For $e : \tau$, define a set function on terminal outcomes by

$$\langle e : \tau \rangle(A) = \lim_{n \to \infty} \nu_n^e(A), \qquad A \in \mathcal{F}_{T_\tau}.$$

Lemma F.44 (Big-step semantics is well-defined). *Conditional on the operational measurability obligations, the set function in Definition F.43 is a subprobability measure and, as a function of its initial expression, a sub-Markov kernel into T_τ.*

Proof. By Lemma F.42, each limit exists and equals a supremum. If $(A_j)_{j \in \mathbb{N}}$ are pairwise disjoint measurable subsets of T_τ, then monotone convergence on $\mathbb{N}$ with counting measure gives

$$\langle e : \tau \rangle\left(\bigcup_j A_j\right) = \lim_n \sum_j \nu_n^e(A_j)$$

$$= \sum_j \lim_n \nu_n^e(A_j) = \sum_j \langle e : \tau \rangle(A_j).$$

Also $\langle e : \tau \rangle(T_\tau) \le 1$. For a fixed measurable terminal event, the finite-step kernel lemma makes its probability measurable in the initial expression, and the pointwise limit of these measurable functions is measurable. □

No coupled finite-step or big-step transfer is claimed here. Such a transfer additionally requires the open weak compiler simulation and a proof that restriction to terminal outcomes commutes with component pushforward.

Proposition F.45 (Conditional closed Boolean semantic preservation). *Let $e : \mathbf{bool} \in Expr^{\cup}_{\mathbf{bool}}$ be closed. Assume a measurable weak target kernel and a measurable weak coupling whose marginals are the source step and target weak step, respectively. Assume also that generated administrative reduction is normalizing, that the terminal restrictions of the coupling iterates have an increasing limit, and that the target projection of this limit is the ordinary target terminal semantics. If related terminal Boolean values agree and the coupling preserves observation-failure mass, then $\langle e : \mathbf{bool} \rangle = \langle \mathcal{D}[\![e : \mathbf{bool}]\!] : \mathbf{bool}$*

```
let x1 = uniform(0,1) in
let x2 = if x1 < 0.56 then uniform(0,2) else uniform(0,3) in
let x3 = if x2 < 0.92 then uniform(0,4) else uniform(0,5) in
let x4 = if x3 < 0.47 then uniform(0,6) else uniform(0,7) in
let x5 = if x4 < 0.51 then uniform(0,8) else uniform(0,9) in
x5 < 0.5
```

Fig. 28. Conditional Independent (Figure 36a): Program of size 5. Each variable depends on its immediate predecessor variable via a conditional branch. Each uniform distribution is distinct.

```
let x1 = uniform(0,1) in
let x2 = if x1 < 0.56 then uniform(0,2) else uniform(0,3) in
let x3 = if x1 < 0.45 then uniform(0,4) else uniform(0,5) in
let x4 = if x1 < 0.18 then uniform(0,6) else uniform(0,7) in
let x5 = if x1 < 0.48 then uniform(0,8) else uniform(0,9) in
x5 < 0.5
```

Fig. 29. Conditional Independent Fork (Figure 36b): Program of size 5. Each variable x_i depends on the same predecessor variable x_1 through a conditional branch. This creates a fork-shaped dependency pattern in which x_1 serves as the sole parent of all subsequent variables. Each uniform distribution is distinct.

Proof. By the terminal-limit compatibility premises, the limiting coupling has the two displayed ordinary terminal semantics as its component pushforwards. Agreement of related terminal Booleans and preservation of observation-failure mass make those pushforwards equal on true, false, and $\bot$.

□

G Benchmarks

G.1 Asymptotic Scaling

G.1.1 Code Sketches for Scaling Benchmarks. This section contains code snippets of the programs used in the scaling experiments in Section 7.2. We implemented a Python script that automatically generates these programs based on configurable parameters such as program size, guard structure, and dependency pattern. For reproducibility, we used a fixed random seed to generate each program. All r_i values in the generated programs, where $0 \le r_i \le 1$ is randomly chosen, are rounded to two decimal places.

All of these programs follow the same structural pattern: they define a sequence of random variables $X_1, \ldots, X_n$, where each X_i is drawn from a uniform distribution whose parameters depend on the value of a single predecessor variable through a conditional branch of the form

$$X_i = \begin{cases} \mathrm{Unif}(a_i, b_i), & X_{\pi(i)} < r_i, \\ \mathrm{Unif}(a_i', b_i'), & \text{otherwise}, \end{cases}$$

for some parent-selection function $\pi : \{2, \ldots, n\} \to \{1, \ldots, i-1\}$. Thus each program is a Bayesian network of conditionally independent nodes, differing only in the choice of $\pi(i)$ that determines the dependency pattern.

The programs in Figure 28 follow the scheme depicted above where the program size is n, i.e., the number of comparisons performed is n. In Figure 29, each x_i does not necessarily depend on x_{i-1} but on some randomly chosen predecessor[3]. In Figures 30 and 31, we additionally choose

[3]We generate these programs in a randomized manner with a fixed seed for reproducibility